



Anti-oxidative stress and gastroprotective effect of Tri-Tharn-Thip tea against ethanol-induced gastric ulcer in rats

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ARTICLE INFO

Article History:

Received 21 March 2024

Revised 14 May 2024

Accepted 15 May 2024

Available online 18 May 2024

Edited by Dr P. Bhattacharyya

Keywords:

Animal model

Anti-gastric ulcer

Antioxidant enzyme

Herbal remedy

Oxidative stress

ABSTRACT

Background: Tri-Tharn-Thip, a traditional Thai medicine, is a herbal formula used to promote milk production, wound healing, and gastric ulcer. However, no previous studies have reported its anti-gastric ulcer activity. This study aimed to evaluate the gastroprotective effects of Tri-Tharn-Thip tea (Tri-TTa) and explore its potential mechanisms of action by measuring the levels of reactive oxygen species (ROS), malondialdehyde (MDA), superoxide dismutase (SOD), catalase (CAT), and glutathione (GSH), mucin, and inducible nitric oxide synthase (iNOS).

Methods: For this purpose, 24 Wistar rats were divided into four groups. Group I, sham group, received distilled water; Group II, vehicle control group, received distilled water; Group III, positive control group, received omeprazole (20 mg/kg/d); and Group IV, Tri-TTa group, received Tri-TTa (500 mg/kg/d). The treatment protocol lasted 7 d, and all groups except for the sham group received 80 % ethanol on the last day, 4 h before scarification. The gastric tissue was removed; the ulcer area, ulcer score, and pH values were evaluated; and a histopathological examination of the gastric tissue was conducted. ROS, MDA, iNOS, SOD, CAT, and GSH levels were determined using ELISA.

Results: Pretreatment with 500 mg/kg Tri-TTa significantly reduced the gastric ulcer area and ROS, MDA, and iNOS levels in rats with ethanol-induced gastric ulcers. Tri-TTa increased the levels of SOD, CAT, and GSH in gastric tissues. Tri-TTa administration also significantly increased mucin content.

Conclusion: Tri-TTa exhibited gastroprotective properties that could be linked to decreased oxidative stress markers and elevated antioxidant enzyme levels.

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1. Introduction

Gastric ulcers (GUs), the primary form of peptic ulcers, represent the most prevalent digestive disorder in the current century (Shareef et al., 2022). Various contributing factors, including exposure to excessive acid, smoking, *Helicobacter pylori* infection, ischemia, hypoxia, NSAID use, and alcohol consumption, may lead to GUs (Gilani et al., 2022). Therapeutic approaches and treatments for peptic ulcers involve various medications, such as prostaglandin analogs, proton

pump inhibitors (omeprazole), histamine receptor blockers (cimetidine, ranitidine, and famotidine), and mucosal blockers (Sofi et al., 2020). Omeprazole, a widely recognized synthetic chemical, is extensively used to treat GUs. Although omeprazole has therapeutic potential in alleviating GU symptoms, it has also been associated with various side effects, such as back, leg, or stomach pain, bleeding or crusting, blisters, cloudy urine, burning and painful urination, and swelling of the lips (Hamzeloo-Moghadam et al., 2021). Hence, the quest for safer and more effective alternatives to therapeutic agents for GUs has become a significant scientific focus in recent years.

The ethanol-induced GU rat model is widely used to study gastroprotective effect of various natural herbal (Simões et al., 2019). Ethanol, a colorless volatile liquid, is a harmful agent associated with numerous pathologies. It is orally administered to experimental animals to induce ulcers and gastric lesions (Jabbar et al., 2023). Ethanol disrupts the integrity of the gastric barrier by inducing cell exfoliation and increasing the mucosal permeability. The influx of white blood cells (WBCs), especially neutrophils, to the damaged site triggers

Abbreviations: Abbreviations: CAT, Catalase; ELISA, Enzyme-linked immunosorbent assay; GA, gastric area; GSH, Glutathione; GUs, Gastric ulcers; iNOS, Inducible nitric oxide synthase; IL-1beta, interleukin-1; IL-6, interleukin-6; MDA, Malondialdehyde; NSAID, Non-steroidal Anti-inflammatory drugs; SOD, Superoxide dismutase; ROS, Reactive oxygen species; TNF-alpha, tumor necrosis factor -alpha; Tri-TTa, Tri-Tharn-Thip tea; UA, Ulcer area; UI, Ulcer index; WBC, white blood cells; MgCl₂, Magnesium chloride

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<https://doi.org/10.1016/j.sajb.2024.05.027>

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increased concentrations of reactive oxygen species (ROS), malondialdehyde (MDA) as a lipid peroxidation marker, and other inflammatory mediators, such as inducible nitric oxide synthase (iNOS), tumor necrosis factor α (TNF- α), interleukin-1 β (IL-1 β), and interleukin-6 (IL-6), resulting in oxidative damage with detrimental effects on gastric cells. Oxidative stress also plays a role in ethanol-induced gastric mucosal injury (Qin et al., 2019).

Herbal bioactive compounds, known for their strong antioxidant, anti-inflammatory, and anti-apoptotic effects, have the potential to provide effective gastrointestinal protection (Raish et al., 2021). Tri-tharn-thip (Tri-TT), a traditional Thai medicine, is an herbal formula used to promote milk production, wound healing effects, and anti-diarrhea effects (Chonsut et al., 2023). The Tri-TT formula is composed of the dried roots of three different herbs: *Cassia fistula*, *Ficus benjamina*, and *Pithecellobium dulce*. Previous studies have reported that Tri-TT shows potential as a promising natural source with antioxidant activity *in vitro*, indicating the presence of polyphenols (Sanpinit et al., 2024). In addition, Chonsut et al. (2023) reported that Tri-TT can boost milk production in a lactating rat model (Wetchakul et al., 2022; Chonsut et al., 2023). Based on the potent antioxidant effects of Tri-TT, this traditional Thai herbal formula may have a favorable therapeutic effect against ethanol-induced GUs. The objective of this study was to evaluate the effects of Tri-TT on ethanol-induced GUs in rats.

2. Methods

2.1. Plant material collection and preparation of dry Tri-Tharn-Thip tea (Tri-TTa) extract

Briefly, the Tri-TTa extract preparation, and plant materials, including the dried root powders of *P. dulce*, *C. fistula*, and *F. benjamina*, were mixed in a 1:1:1 ratio. Three grams of Tri-TTa infusions were created by placing one tea bag (size 7 × 9 cm) into 100 mL of freshly boiled water at a temperature of 98 ± 2 °C and soaking for 3 min without stirring. The Tri-TTa extract was filtered through Whatman No. 1 filter paper. Subsequently, the extracts were completely dried in a lyophilizer and kept at -20 °C until subsequent use.

2.2. Animals and experimental protocol

Male Wistar rats, weighing 150–200 g, were obtained from Nomura Siam International Co., Ltd. They were acclimatized under standard laboratory conditions (22 ± 2 °C, 55 ± 10 humidity and 12 h light-dark cycle), with free access to water *ad libitum* and standard diet for 7 d within the Laboratory Animal Unit of the Research Institute for Health Sciences of Walailak University. All the protocols were approved by the Animal Ethics Committee of Walailak University, Thailand (WU-ACUC-65043).

2.3. Ethanol induced gastric ulcer model

The rats were randomly divided into four groups, with six rats in each group, as follows:

- 1) The sham (normal) group received distilled water
- 2) The vehicle control group received distilled water
- 3) The positive group received omeprazole (20 mg/kg/d)
- 4) The Tri-TTa group received Tri-TTa (500 mg/kg/d)

All rats were treated orally for seven consecutive days. On day six, the animals were fasted for 24 h with free access to drinking water. On the last day, GUs were induced by the oral administration of 80 % ethanol solution to the animals. Four hours after induction, the

animals were euthanized using an overdose of thiopental sodium. Blood samples were collected and the stomachs were dissected.

2.4. Determination of the pH in gastric juice

The stomachs were opened along the greater curvature, and pH indicator strips were inserted into the gastric juice. The stomachs were thoroughly rinsed with 0.9 % normal saline, and the wet weights of the tissues were recorded.

2.5. Gross evaluations of stomachs

After weighing the stomach tissue, the samples were placed on plastic plates and fastened to a pin. Inward part of the inner stomach was photographed using a camera. The total gastric area (GA) and total GU area (UA) (cm²) were determined using Image J™ image processing software. The ulcer index (UI) was calculated using the following formula:

UI = Sum of the total score/Number of ulcers

Scoring of ulcers

Normal stomach 0

Red coloration 0.5

Spot ulcer 1

Hemorrhagic streak 1.5

Deep ulcer 2

Perforation 3

The stomach was divided into three sections. The first portion of the stomach was immersed in 1 % Alcian blue solution to determine the mucin content, the second portion was kept at -80 °C for evaluation of gastric biochemical parameters, and the last portion was fixed in 10 % formalin solution for histological analysis.

2.6. Determination of mucin content

The first portion of the gastric tissue was immersed in a 1 % Alcian blue solution prepared in 50 mM sucrose and 0.16 mM sodium acetate (pH 5) for 2 h at room temperature to determine mucin content. Subsequently, the gastric tissue was washed 2 times (for 15 and 45 min) with 0.25 M sucrose solution to remove the uncomplexed dye and was separately incubated in 0.5 M MgCl₂ for 2 h at room temperature while shaking at 30-min intervals to elute the dye complexed with mucous of the gastric. After 2 h, 5 mL of the complex dye (blue solution) was added to 5 mL of diethyl ether. The resulting emulsion was centrifuged at 4000 rpm for 10 min at room temperature, and the absorbance of the aqueous layer was recorded at 605 nm using a microplate reader. The concentration of Alcian blue was calculated via linear regression using a calibration curve obtained from standard serial dilutions of Alcian blue at different concentrations. The results are expressed in milligrams of Alcian blue/g of tissue.

2.7. Blood collection to evaluate the number of inflammatory cells

Blood samples were collected in ethylenediaminetetraacetic acid-containing tubes to evaluate the WBC count and percentage of WBC types, including neutrophils, lymphocytes, eosinophils, basophils, and monocytes.

2.8. Gastric tissue preparation to evaluate the gastric biochemical parameters

Gastric tissues were homogenized in cold phosphate buffer (pH 7.4) and centrifuged at 3000 rpm for 20 min, and the supernatants were used to estimate oxidative stress markers, antioxidant enzymes, and inflammatory cytokines.

2.9. Determination of oxidative stress markers

ROS and MDA concentrations in gastric homogenates were quantified using standard ELISA kits (Sunlong Biotech Co., Ltd., Hangzhou, China). The absorbance was measured at 450 nm using a microplate reader.

2.10. Determination of antioxidant enzymes

Superoxide dismutase (SOD), catalase (CAT), and glutathione (GSH) concentrations in gastric homogenates were quantified using commercially available standard ELISA kits (Sunlong Biotech Co., Ltd., Hangzhou, China). The absorbance was measured at 450 nm using a microplate reader.

2.11. Determination of inflammatory cytokines

iNOS concentrations in gastric homogenates were quantified using standard ELISA kits (Sunlong Biotech Co., Ltd., Hangzhou, China). The absorbance was measured at 450 nm using a microplate reader.

2.12. Hematoxylin and eosin (H&E) staining of gastric tissue

The last portion of the gastric tissue was fixed in 10 % formalin solution, dehydrated by immersing in 70–100 % ethanol solution, and then embedded in paraffin. The stomach tissue was sectioned into 5- μ m thickness and stained with H&E. Tissue sections were examined under a light microscope at 10X for pathological changes, such as submucosal edema, vasodilation, and inflammatory cell infiltration. All slides were photographed under a microscope.

2.13. Statistical analysis

One-way analysis of variance and Duncan's post-hoc test were used to identify significant differences between the vehicle and treatment groups (SPSS version 20). Statistical significance was set at $P < 0.05$. All data are reported as mean $\pm$ SEM.

3. Results

3.1. Effect of Tri-TTa on ulcer area and ulcer index in ethanol-induced GU rats

The results indicated significant higher ulcer area in the vehicle group (0.59 ± 0.03 cm²) than that in the sham group. Pre-treatment with 500 mg/kg of Tri-TTa resulted in a decrease in ulcer area (0.02 ± 0.01 cm²) compared to the vehicle group. Similarly, pre-treatment with 20 mg/kg of omeprazole (0.14 ± 0.01 cm²) resulted in the suppression of ulcer area compared to the vehicle group. The ulcer index also demonstrated a significant decrease in both the 500 mg/kg Tri-TTa and 20 mg/kg omeprazole groups compared to that in the vehicle group (Fig. 1 and Table 1).

3.2. Effect of Tri-TTa on mucin content and pH values in ethanol-induced GU rats

The results of ethanol-induced GU showed a significant decrease of mucin level (18.88 ± 2.77 μ g Alcian blue/g of tissue) in gastric tissue when compared to the sham group (29.19 ± 1.02 μ g Alcian blue/g of tissue). However, the rats pretreated with omeprazole (32.02 ± 1.53 μ g Alcian blue/g of tissue) and Tri-TTa (32.10 ± 1.70 μ g Alcian blue/g of tissue) showed significant increase in mucin level when compared with the vehicle group. No significant differences were observed between the omeprazole and Tri-TTa groups. The pH values

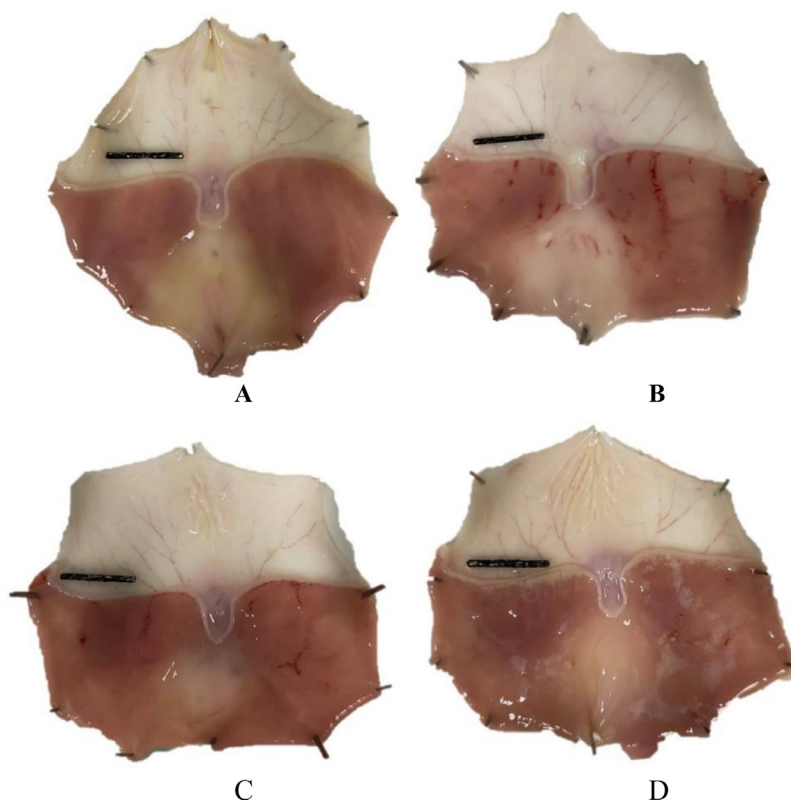


Fig. 1. Effects of omeprazole and Tri-Tharn-Thip tea (Tri-TTa) on the gross appearance of the gastric mucosa in ethanol-induced gastric ulcers in rats. (A) Sham group; (B) vehicle group; (C) OMZ group (20 mg/kg); (D) Tri-TT (500 mg/kg) group. Scale bar = 1 cm.

Table 1

Effect of Tri-Tharn-Thip tea (Tri-TTa) on gastric lesions induced by ethanol induced acute gastric lesions in rats.

Group	GA (cm ²)	UA (cm ²)	UI
Sham	15.60 ± 0.10	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a
Vehicle	15.92 ± 0.25	0.59 ± 0.03 ^{#,b}	1.12 ± 0.05 ^{#,d}
Omz.20	15.79 ± 0.37	0.14 ± 0.01 ^{*,a}	0.96 ± 0.10 ^{*,bc}
Tri-TTa	15.62 ± 0.12	0.02 ± 0.01 ^{*,a}	0.69 ± 0.10 ^{*,b}

Data were expressed as mean ± SEM (n = 6) and analyzed using one-way ANOVA and post hoc Duncan multiple comparisons test.

* Significantly different from vehicle control at $p < 0.05$.

Significantly different from sham group at $p < 0.05$. a-d Values in the same column with different superscripts are significantly different ($p < 0.05$). UA; Total ulcer area, GA; Total gastric area UI; ulcer index, % UA; % of ulcerated area, Omz.; Omeprazole (20 mg/kg), Tri-TTa; Tri-Tharn-Thip tea (500 mg/kg).

of the gastric juice were not significantly different between ethanol-induced GU and normal rats (sham group). However, pH was significantly enhanced in rats pretreated with omeprazole and Tri-TTa. (Table 2)

3.3. Effect of Tri-TTa on ROS and MDA in ethanol-induced GU rats

There was a significant increase in ROS and MDA levels in the gastric tissue of ethanol-induced GU rats (669.44 ± 68.58 pg/mL and 103.12 ± 13.98 pg/mL, respectively) compared to the sham group (513.07 ± 32.22 pg/mL and 62.63 ± 1.59 pg/mL, respectively). In contrast, the levels of ROS and MDA were significantly decreased in rats pretreated with omeprazole and Tri-TTa compared to those in the vehicle group, as shown in Fig. 2A and B.

3.4. Effect of Tri-TTa on SOD, CAT, and GSH in ethanol-induced GU rats

The results showed that the levels of SOD and GSH in the gastric tissue were significantly decreased in ethanol-induced GU rats (368.43 ± 78.38 pg/mL and 103.26 ± 6.54 ng/L, respectively) compared with those in the sham group (893.06 ± 68.07 pg/mL and 153.46 ± 5.93 ng/L, respectively). However, the rats pretreated with omeprazole (702.05 ± 52.41 pg/mL and 156.85 ± 5.44 ng/L, respectively) and Tri-TTa (732.67 ± 42.22 pg/mL and 141.76 ± 6.89 ng/L, respectively) showed significant increase in SOD and GSH level when compared with the vehicle group. Furthermore, no significant differences were observed among the omeprazole, Tri-TTa, and sham groups (Fig. 3A and C). CAT levels in the vehicle group were lower than those in the sham group; however, the difference was not significant. Conversely, pre-treatment of rats with omeprazole (17.48 ± 0.60 pg/mL) and Tri-TTa (15.47 ± 0.47 pg/mL) led to a significantly higher CAT level than in the vehicle group (13.53 ± 0.46 pg/mL) (Fig. 3B).

Table 2

Effect of Tri-Tharn-Thip tea (Tri-TTa) on gastric parameters induced by ethanol induced acute gastric lesions in rats.

Group	Mucin content (μ g Alcian blue/g of tissue)	pH value
Sham	29.19 ± 1.02 ^b	2.67 ± 0.21 ^a
Vehicle	18.88 ± 2.77 ^{*,a}	2.17 ± 0.17 ^a
Omz.20	32.02 ± 1.53 ^{*,b}	5.17 ± 0.70 ^{*,b}
Tri-TTa	32.10 ± 1.70 ^{*,b}	5.25 ± 0.75 ^{*,b}

Data were expressed as mean ± SEM (n = 6) and analyzed using one-way ANOVA and post hoc Duncan multiple comparisons test.

* Significantly different from vehicle control at $p < 0.05$.

Significantly different from sham group at $p < 0.05$.

a-b Values in the same column with different superscripts are significantly different ($p < 0.05$). UA; Total ulcer area, GA; Total gastric area UI; ulcer index, %UA; % of ulcerated area, Omz.; Omeprazole (20 mg/kg), Tri-TTa; Tri-Tharn-Thip tea (500 mg/kg).

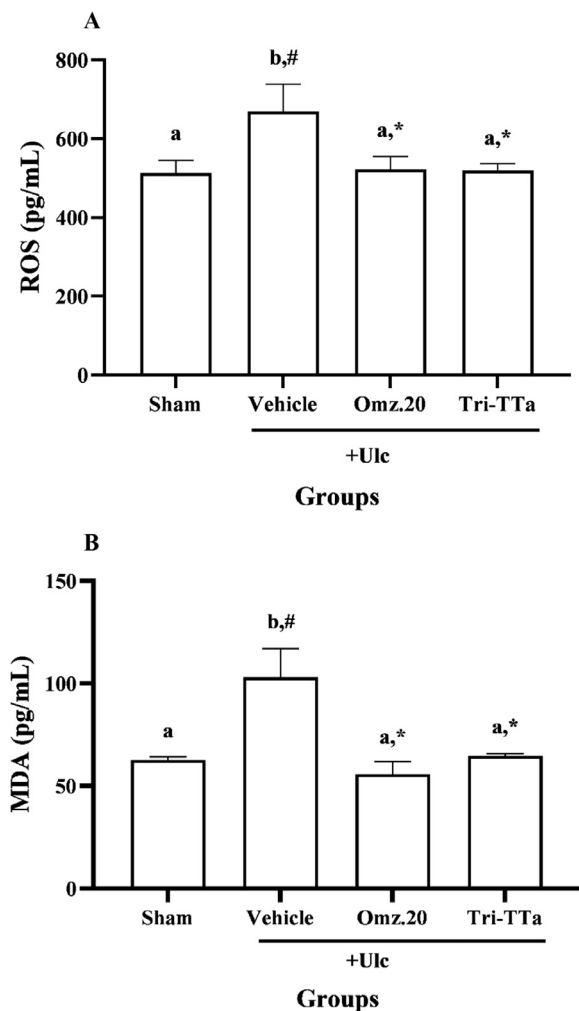


Fig. 2. Effects of omeprazole and Tri-Tharn-Thip tea (Tri-TTa) on the ROS (A) and MDA (B) levels in the gastric of rats exposed to ethanol. Data are expressed as the mean SEM (n = 6 per group) and were analyzed using one-way ANOVA and Duncan's post hoc multiple-comparison test. * Significantly different from the vehicle control at $p < 0.05$. # Significantly different from the normal group at $p < 0.05$. Different letters (a-b) indicate significant differences ($p < 0.05$) between treatments determined by Duncan's multiple-range test.

3.5. Effect of Tri-TTa on iNOS in ethanol-induced GU rats

The results showed that the levels of iNOS in the gastric tissue were significantly increased in ethanol-induced GU rats (6.76 ± 0.61 ng/mL) compared with those in the sham group (4.83 ± 0.56 ng/mL). However, the rats pretreated with omeprazole (4.45 ± 0.65 ng/mL) and Tri-TTa (4.85 ± 0.07 ng/mL) showed significant decrease in iNOS level when compared with the vehicle group (Fig. 4).

3.6. Effect of Tri-TTa on the white blood cell (WBC) count and percentage of WBC types in ethanol induced gastric ulcer rat

There was a significant increase in WBC, neutrophil, and lymphocyte counts in the blood samples of ethanol-induced GU rats compared with those in the sham group. However, rats pretreated with omeprazole and Tri-TTa showed a significant decrease in WBC, neutrophil, and lymphocyte counts compared to those in the vehicle group. Eosinophils, basophils, and monocytes were not detected in any rat (Table 3).

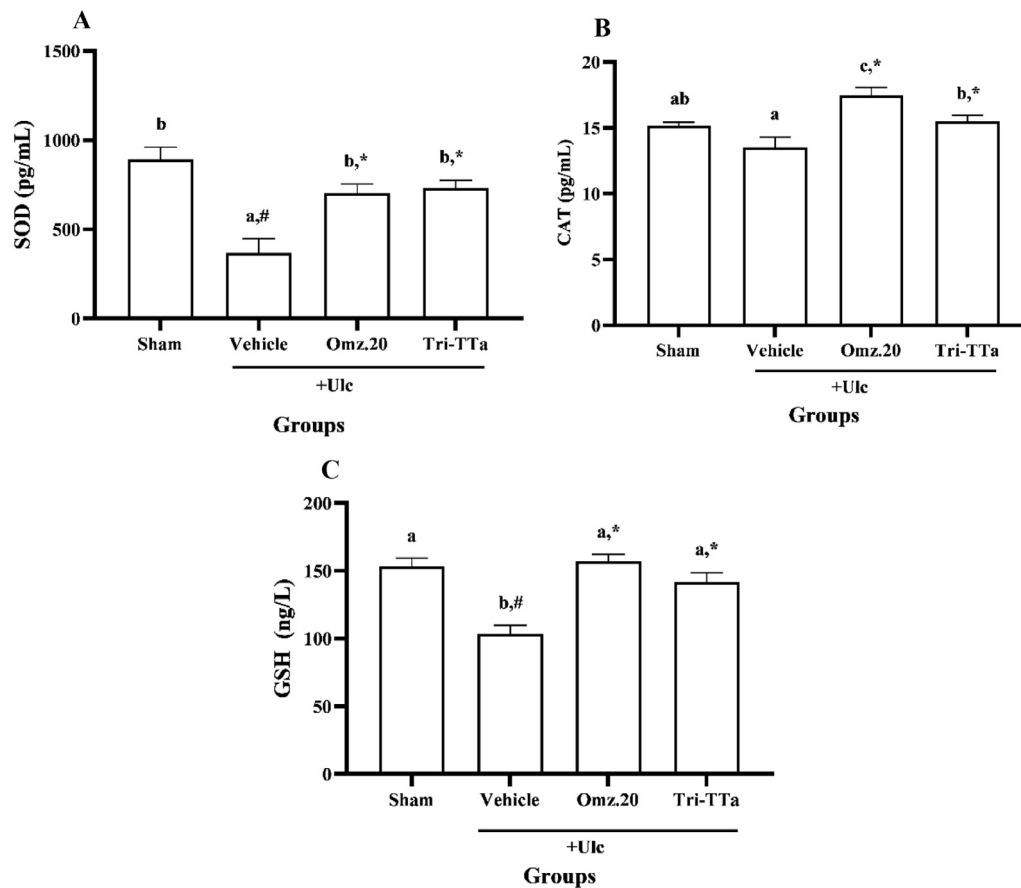


Fig. 3. Effects of omeprazole and Tri-Tharn-Thip tea (Tri-TTa) on the levels of SOD (A), CAT (B), and GSH (C) in the gastric of rats exposed to ethanol. Data are expressed as the mean $\pm$ SEM ($n = 6$ per group) and were analyzed using one-way ANOVA and Duncan's post hoc multiple-comparison test. * Significantly different from the vehicle control at $p < 0.05$. # Significantly different from the normal group at $p < 0.05$. Different letters (a–c) indicate significant differences ($p < 0.05$) between treatments determined by Duncan's multiple-range test.

3.7. Effect of Tri-TTa on the histopathological examination of the gastric tissue in ethanol-induced GU rats

H&E staining was performed. The results revealed that the gastric tissue of the sham group (Fig. 5A) exhibited normal architecture of

the gastric mucosal wall without any signs of deterioration. In contrast, gastric tissue in the vehicle group showed a significant area of mucosal erosion, vasodilation, extensive edema, and infiltration of inflammatory cells into the submucosal layer (Fig. 5B). Remarkably, gastric tissues pretreated with omeprazole and Tri-TTa showed a clear alleviation of the severity of mucosal lesions, decreased edema, and reduced infiltration of inflammatory cells into the submucosal layer (Fig. 5C and D).

4. Discussion

An animal model of ethanol-induced GU is considered a reliable tool for investigating new drugs with anti-GU properties (Oates and Hakkinen, 1988). The administration of ethanol leads to gastric mucosal damage and subsequent inflammatory cell infiltration, reduces the production of gastric mucus, and decreases gastric juice pH (Rahman et al., 2020). In the current study, oral administration of 80 % ethanol caused a GU with an ulcer index score of 1.12. This indicates that the severity of the gastric lesions is characterized by a spot GU and a nearly hemorrhagic streak. Remarkably, pretreatment with omeprazole (20 mg/kg) and Tri-TTa (500 mg/kg) resulted in a reduction of the ulcer index score to 0.96 and 0.69, respectively, suggesting that the gastric lesions in both groups demonstrated a milder severity, characterized mainly by red-colored areas rather than pronounced ulceration. The entire gastric mucosa is coated with a protective mucous layer, with mucin glycoproteins constituting the primary components (Byrd and Bresalier, 2000). Several studies have reported that ethanol-induced GUs lead to a reduction in the mucin content and pH within the gastric tissue (Kim et al., 2020; Rubab et

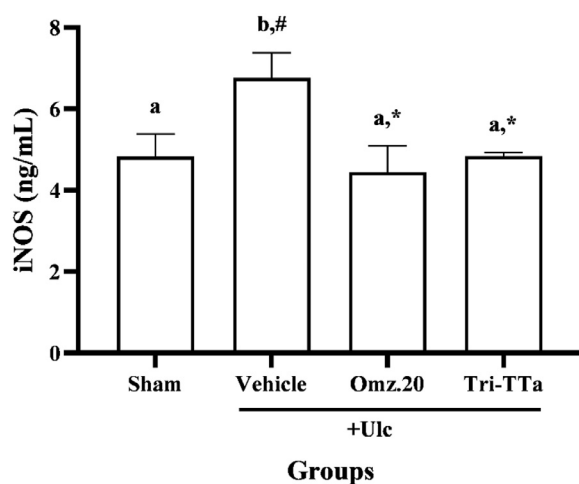


Fig. 4. Effects of omeprazole and Tri-Tharn-Thip tea (Tri-TTa) on the levels of iNOS in the gastric of rats exposed to ethanol. Data are expressed as the mean $\pm$ SEM ($n = 6$ per group) and were analyzed using one-way ANOVA and Duncan's post hoc multiple comparisons test. *Significantly different from the vehicle control at $p < 0.05$. #Significantly different from the normal group at $p < 0.05$. Different letters (a–b) indicate significant differences ($p < 0.05$) between treatments determined by Duncan's multiple range test.

Table 3
Effect of Tri-Tharn-Thip tea (Tri-TTa) on the white blood cell (WBC) count and percentage of WBC types in blood of rats induced by ethanol induced acute gastric lesions in rats.

Group/ Parameters	Sham	Vehicle	Omz.20	Tri-TTa
WBC (cells/mm ³)	3267 ± 49.44 ^a	6283 ± 90.98 ^d	5483 ± 101.38 ^c	4250 ± 220.23 ^b
Neutrophil (%)	11.00 ± 0.60 ^a	16.00 ± 1.43 ^b	11.00 ± 0.67 ^a	10.00 ± 0.71 ^a
Lymphocyte (%)	74.00 ± 1.08 ^a	87.00 ± 0.70 ^b	74.00 ± 0.88 ^a	76.00 ± 2.69 ^a
Eosinophils (%)	0.00 ± 0.00 ^a	0.33 ± 0.21 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a
Basophils (%)	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a
Monocytes (%)	0.00 ± 0.00 ^a	0.00 ± 0.00	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a

Data were expressed as mean ± SEM (n = 6) and analyzed using one-way ANOVA and post hoc Duncan multiple comparisons test. Different letters (a-d) indicate the same row with different superscript letters are significantly different (p < 0.05).

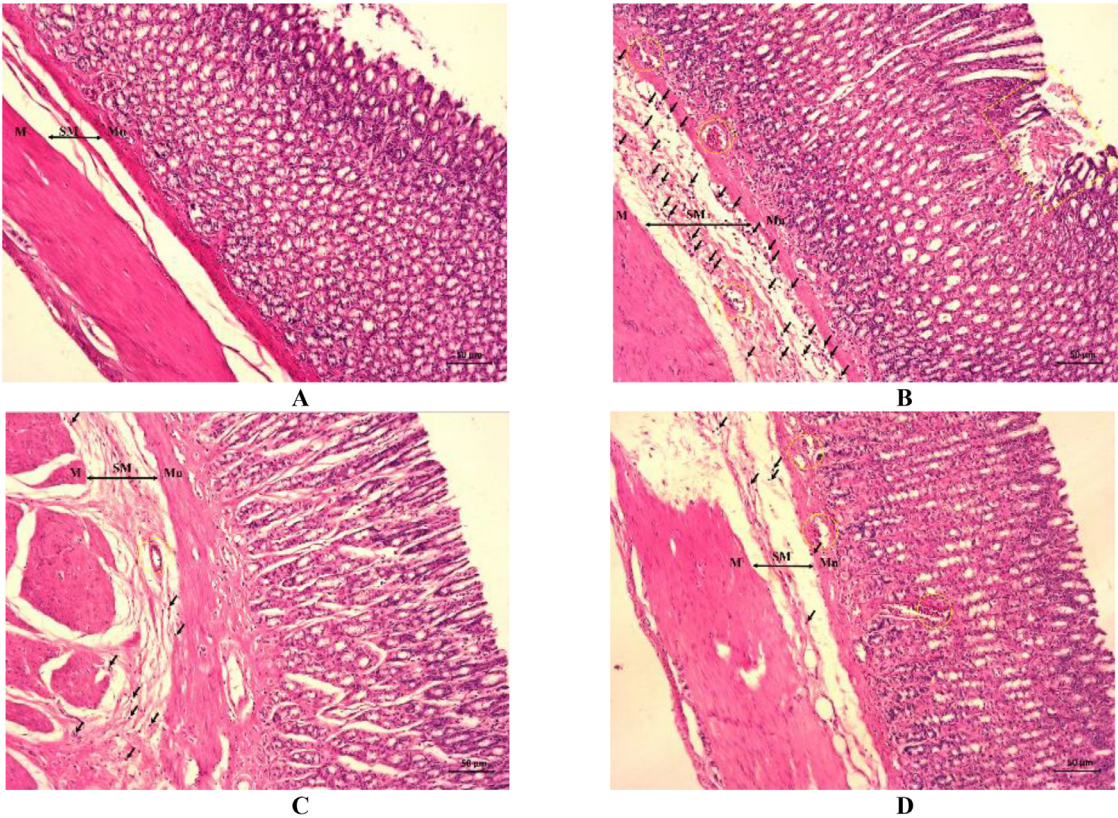


Fig. 5. Histological assessment of the effects of omeprazole and Tri-TTa on ethanol induced acute gastric lesions in rats. (A) Sham group; (B) vehicle group; (C) OMZ group (20 mg/kg); (D) Tri-TT (500 mg/kg) group. M = muscle layer, SM = submucosa layer, Mu = mucosa layer, yellow circle = blood vessel, black arrow (↓) = the inflammatory cells present in the submucosa layer and yellow square = ulcer area. (scale bar = 50 μm) (Magnification ×10).

al., 2022; Sanpinit et al., 2023). In agreement with previous studies, the present findings showed that ethanol administration in the vehicle group led to a significant decrease in mucin content and pH in the gastric tissue. However, pretreatment of rats with omeprazole (20 mg/kg) and Tri-TTa (500 mg/kg) significantly increased mucin content and pH value. The mucosal layer protects the gastric mucosa against ethanol-induced GUs (Rahman et al., 2020). Numerous studies have reported the properties of natural herbs and their phytoconstituents, which enhance mucus production, leading to a reduction in gastric area and UI in an ethanol-induced model (Tijani et al., 2022; Yoo et al., 2018; Gugliandolo et al., 2021; Al-Rejaie et al., 2012). Therefore, the significant anti-GU activity of Tri-TTa could be due, in part, to its ability to stimulate mucin production and reduce the acidity of gastric juice. An imbalance between aggressive factors, such as oxidative stress, and protective factors, such as antioxidant enzymes, promotes GU formation (Kim et al., 2020). The ethanol-induced GU model results in oxidative stress by increasing generation of ROS, increased production of MDA, elevated levels of inflammatory

cytokines, such as iNOS, TNF-alpha, IL-1beta, and IL-6, as well as a decrease in the production of antioxidant enzymes, such as CAT, SOD, and GSH (Raish et al., 2021). In our study, we found significantly higher levels of ROS, MDA, and iNOS in ethanol-induced gastric tissues in the vehicle group. Moreover, the levels of CAT, SOD, and GSH were significantly decreased in ethanol-induced gastric tissue. In contrast, pretreatment with omeprazole (20 mg/kg) and Tri-TTa (500 mg/kg) significantly reduced the ROS, MDA, and iNOS levels. In addition, pretreatment with omeprazole and Tri-TTa significantly increased CAT, SOD, and GSH levels. These results suggest that Tri-TTa exerts its gastroprotective effects by reducing oxidative stress and inducing the production of antioxidant enzymes. Tri-TTa comprises herbal components, such as *C. fistula*, *P. dulce*, and *F. benjamina*. However, scientific evidence regarding the gastroprotective effects of these herbs on the roots is lacking. Wetchakul et al. (2022) and Sanpinit et al. (2024) reported that the chemical constituents of Tri-TTa include phenolic acid groups, such as galactaric acid, quinic acid, chebulic acid, glucogallic acid, fukiic acid, and 2-nitroanisoole, and

flavonoid groups, such as hydroquinone, gentisic acid, quercetagitrin, chlorogenic acid, epicatechin, and diospyrin, which exhibit potent antioxidant activities by various methods. Polyphenols exhibit various biological functions, including antioxidant and gastroprotective effects (Sumbul et al., 2011; Chiu et al., 2021). Our results suggest that the gastroprotective effects of Tri-TTa may be attributed to its chemical components, which enhance antioxidant enzymes and reduce oxidative stress.

According to our results, Tri-TTa exhibited gastroprotective effects against ethanol-induced GUs. These results suggest that Tri-TTa is a powerful remedy for protecting gastric mucosal lesions by inhibiting oxidative activity, including the inhibition of lipid peroxidation and stimulation of antioxidant enzymes. However, further studies are needed to determine the exact mechanisms underlying the gastroprotective activity of Tri-TTa.

Ethical statement

This research was approved by the Animal Ethical Committee of Walailak University, Thailand (WU-ACUC-65043) to ensure ethical treatment of the animals.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

CRediT authorship contribution statement

Palika Wetchakul: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Soiphet Net-Anong:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Jo Aan Goon:** Writing – original draft, Writing – review & editing. **Sineenart Sanpinit:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Acknowledgements

The authors would like to thank the Research Institute for Health Science, Walailak University, for the grants.

Funding

This work was supported by the, Research Institute for Health Science, Walailak University, grant number WU-IRG-65-020.

Data availability

The data associated with this study can be made available upon reasonable request to the corresponding author.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.sajb.2024.05.027.

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